

Vitor Fernandes

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Resume

I am a PhD student at the University of Minho. My research focuses on the semantics of programming languages with concurrent, probabilistic, and/or quantum features. I enjoy implementing these languages in Haskell to ease the reasoning about programs. The languages I study have an imperative flavor, with a syntax similar to C. I also taught the C language as an invited professor at the University of Minho. Since my MSc. I have been using Git for project management. I am eager to learn new programming languages and to apply the knowledge obtained in the development of useful quantum algorithms.

Education

- 🎓 **University of Minho** – PhD student in Computer Science with Professor Luís S. Barbosa, Professor Renato Neves, and Professor Benoît Valiron Feb 2021 - Present
- 🎓 **Université Paris Saclay (Laboratoire Méthodes Formelles)** – Visiting PhD. Student Feb 2024 - Aug 2024
- 🎓 **University of Minho** – BSc. & MSc. Engineering Physics Sep 2014 - Dec 2019

Experience

- 👤 **University of Minho**, Invited Teacher Feb 2025 - Jul 2025
 - Imperative Programming (C), Quantum Logic (ZX-Calculus)
- 📖 **Supervision MSc. thesis:** An interpreter for a concurrent quantum language Sep 2022 - Mar 2024

Skills

- 🛠️ **Programming Languages:** Haskell, C, Wolfram Mathematica, Matlab
- 🛠️ **Project Management:** Git/GitHub

Project

- 🔗 **qProbConc:** Haskell interpreter of a concurrent, probabilistic, and quantum language 2025

Publications

- 📄 **Exploring a Quantum Programming Language with Concurrency** (Co-author) 2025
- 📄 **Non-deterministic, Probabilistic, and Quantum Effects Through the Lens of Event Structures** 2024
 - Oral presentation at the *APLAS24* conference
- 📄 **Quantum simulation of the ground-state Stark effect in small molecules: a case study using IBM Q** 2021
 - Published in the *Soft Computing* journal

Award

- 🏆 **Academic Merit Scholarship:** corresponding to the scholar year 2018/2019, given by the Portuguese Ministry of Science, Technology and Higher Education 2020
- 🏆 **Excellence award**, best student of the 2nd year of MSc. 2020

Events Attended

- ❄️ **IQM** – Quantum School, Online Dec 2025
- ❄️ **Linear Logic** – Winter School, Centre International de Rencontres Mathématiques Jan 2022
- 🌿 **IBM** – QuantaLab School, International Iberian Nanotechnology Laboratory Oct 2018

Extra-curricular activities

- ⚽ Captain of a football team (Realense Futebol Clube) – **Leadership**
- 🔗 Founder of quantUM seminars (2022) – **Self-initiative**